



NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Batch Number	DB-12
Team Members	B. Hemanth Kumar (23475A0524) G. Nithin (23475A0520) K. Arjun Reddy (22471A05N1)
Guide	T G Ramnadh Babu M. Tech
Title	A Text-to-Speech Dataset for Multimodal Synthetic Media Detection
Domain/Technology	DEEP LEARNING
Base Paper Link	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10124769
Dataset Link	https://zenodo.org/records/6560159
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	With the advancement of deep learning, generating falsified multimedia content has become increasingly accessible, posing serious challenges for media authentication. Existing forensic methods often focus on a single modality (audio or video), which limits their ability to detect sophisticated deepfakes. To address this gap, we propose a pipeline for generating multimodal deepfake datasets by synthesizing speech content using advanced Text-to-Speech (TTS) and Dynamic Time Warping (DTW) techniques. This approach enables the creation of TIMIT-TTS, a synthetic audio dataset featuring 12 state-of-the-art TTS systems and nearly 80,000 tracks. The dataset supports both standalone audio and multimodal research, improving deepfake detection. We conducted experiments in mono- and multimodal settings, highlighting the benefits of cross-modal analysis. Additionally, we enhanced the original detection model and achieved an improved accuracy of 87.63%, demonstrating the effectiveness of multimodal approaches in forensic detection.

Signature of the student(s)

Signature of the Guide

Signature of the project coordinator